Mississippi EVSE Zone Explorer

A statewide ArcGIS Experience Builder application that maps EV charging eligibility zones, utility territories, and equity priorities across Mississippi's federally designated corridors

---

The Problem

Before this application, site developers and planners had no easy way to verify whether a candidate property satisfied the federal NEVI program's core spatial requirements, sitting within one travel-mile of an interchange along a designated corridor, and helping close gaps exceeding 50 miles between compliant chargers across the state's interstate system. Checking either of those rules meant working separately with federal corridor designations, utility service territory maps, and equity screening data, none of it available in one place. For an applicant preparing a grant submission, or a utility trying to understand which sites fell in its own service territory, that meant piecing together several independent data sources before a single candidate site could even be evaluated.

My Approach

I built this as a public-facing **ArcGIS Experience Builder application**, so developers, utilities, and planning staff could evaluate candidate sites against the program's actual eligibility criteria themselves.

Eligibility zones based on real road travel. Candidate zones around each qualifying interchange are built from a network routing analysis along public roads, one travel-mile from the interchange ramp, rather than a simple circular radius, so the boundary reflects the distance a vehicle would actually need to travel, not straight-line distance.

Coverage gaps computed directly against the compliance standard. A dedicated overlay identifies corridor stretches exceeding 50 miles without a station meeting the program's four-port, 150 kW simultaneous output requirement, so the specific stretches of interstate still needing infrastructure are visible on the map rather than buried in a report.

Utility and equity context built in, not separate lookups. Electric utility service territories, covering both electric power associations and investor-owned utilities, are shown directly on the map, so a developer can immediately see which provider to contact for a given site. Federal disadvantaged community designations are layered in as well, since they factor directly into the program's equity investment goals.

Navigation built around distinct tasks. Rather than one dense interface trying to do everything at once, the application separates the main map, the layer control panel, and a dedicated data download page into distinct views, so a user looking to explore layers and a user looking to grab the underlying GIS data each land on the interface built for what they're actually trying to do.

Data made available for download, not just display. The underlying candidate zone, corridor, and utility layers can be downloaded directly from the application as GIS-ready files, so a developer or planner who needs to bring the data into their own software isn't limited to what's visible in the map view.

Technical Highlights

- **Platform:** ArcGIS Experience Builder, built on ArcGIS Online hosted feature layers, with dedicated map, layer, and download views
- **Data sources:** FHWA and MDOT alternative fuel corridor designations, US Department of Energy Alternative Fuels Data Center for existing chargers, Mississippi Public Service Commission and electric power association utility territories, federal CEJST disadvantaged community layer
- **Spatial analysis:** Network drive-distance polygons for interchange eligibility zones, corridor-wide gap analysis against the program's spacing and power output requirements
- **Interactivity:** Multi-view navigation (map, layers, downloads), toggleable utility and equity layers, geocoded search, detailed feature pop-ups for interchanges, chargers, and utility territories
- **Scope:** Statewide coverage of Mississippi's designated alternative fuel corridors across six interstates, supporting a $50.5 million federal allocation across five fiscal years

Why It Matters

A statewide infrastructure program depends on developers and utilities being able to evaluate a candidate site against every relevant criterion at once, proximity, corridor coverage, the right utility contact, and equity priorities, rather than reassembling that picture from separate sources for every site under consideration. Serving as the official spatial reference for a $50.5 million procurement effort means the application's eligibility logic has to be dependable, and building it around the same network-distance methodology the program's rules are actually based on keeps what a user sees on the map consistent with what the program will actually approve.